CASE 1 – Hypoglycemia

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Objectives:

SOM.MK.I.BPM2.3.DM.2.BCHM.1378	Describe the steps of fatty acid β-oxidation, explain the role of the carnitine shuttle in mitochondrial fatty acid transport, and discuss the importance of carnitine in ATP generation from long-chain fatty acids.
SOM.MK.I.BPM2.3.DM.2.BCHM.1380	Describe the metabolic consequences of carnitine deficiency, including changes in energy metabolism and serum fuel profiles during fed and fasting states.
SOM.MK.I.BPM2.3.DM.2.BCHM.1384	Summarize the key steps, cellular locations, and regulation of fatty acid synthesis, β-oxidation, and ketogenesis.
SOM.MK.I.BPM2.3.DM.2.BCHM.1385	Describe the hormonal regulation of blood glucose homeostasis during feeding and fasting, focusing on the roles of insulin and glucagon.
SOM.MK.I.BPM2.3.DM.2.BCHM.1386	Describe the steps and regulation of glycogen synthesis and degradation, and compare glycogen metabolism between the liver and skeletal muscle.
SOM.MK.I.BPM2.3.DM.2.BCHM.1391	Describe the tissue-specific regulation of glycolysis in the liver and skeletal muscle, and integrate key aspects of carbohydrate metabolism including signal transduction, enzyme regulation, glycolysis, gluconeogenesis, glycogen metabolism, glycogen storage disorders, and blood glucose control.

Case:

An 18-month old baby girl to immigrant parents was brought to the emergency room by an ambulance with a 24-hour history of being unwell, 18-hour of persistent diarrhea and fever. Her mother was worried because she was 'not acting like herself', she was not playing, and she slept all the time. There were no contacts with the sick. She has been an otherwise well child with no developmental concerns. She has a positive family history of sudden infant syndrome as her older brother was found dead in his crib after a similar episode at 12 months of age. Her birth history is unremarkable other than the fact that she was born abroad.

Physical examination showed a lethargic infant with marked cold extremities. Her temperature was 35.9°C (96.62°F). Abdominal examination showed hepatomegaly with no abdominal distension. Throat examination showed a hyperemic, exudative pharynx. Urgent capillary blood glucose was 34 mg/dL (documented hypoglycemia).

She was started on immediate treatment and was given two boluses of 2 mL/kg of 10% dextrose solution through ultrasound-guided cannulation. Blood samples were taken for hypoglycemia screening and partial septic screening including liver, renal, and bone profile. Her laboratory findings show the following:

Test	**Value (Normal Range)**	**Test**	**Values (Normal Range)**
<u>Arterial Blood:</u>		**<u>Plasma:</u>**	
pH	7.28	**Cortisol**	

pO_2 **pCO_2** **HCO_3^-** **<u>Venous Blood:</u>** **Lactate**	43 mmHg 32 mmHg 18 mEq/L 2 mmol/L (0.5 – 1)	 **C-peptide** **Insulin**	4354 (Significant adrenal response) 1.73 ng/mL (0.5 – 2.0 ng/mL) 20 mIU/L (< 25 fasting)
<u>Electrolytes:</u> **Sodium** **Potassium** **Chloride** **<u>LFTs:</u>** **Alanine Transaminase** **Aspartate Transaminase** **Alkaline phosphatase**	 135 mEq/L 4.4 mEq/L 110 mEq/L 125 U/L (8 – 20) 86 U/L (8 – 20) 63 U/L (20 – 70)	**Blood Ammonia** **Serum Uric Acid** **C-Reactive Protein** **Total WBC Count**	180 μ/dL (15 to 45 μ/dL) 9.3 mg/dL (2.6 – 7.2) 173 mg/dL (< 3.0) 27,500/microliter (with Neutrophilia)
Free Carnitine **Acylcarnitine**	Reduced Elevated	**Urinary Dicarboxylic Acids** **Acyl-glycine conjugates** **Ketones**	Elevated Elevated Absent

Q1) Give a biochemical and physiological explanation to the lethargy displayed by the infant? (In relation to the plasma glucose levels; Discuss the importance of glucose)

Q2) What is the most likely diagnosis? What could have triggered the hypoglycemia in this infant? Correlate with the plasma cortisol and plasma insulin levels of the patient

Q3) Name several metabolic disorders that can lead to hypoglycemia in childhood.

Q4) What is the most likely diagnosis? What is the function of the deficient enzyme? (Discuss in detail the process of β-oxidation)

Q5) What is the differential diagnosis for hypoketotic hypoglycemia? Why was the patient hypoketotic? (Review ketogenesis, its site of synthesis and ketone body utilization; additionally, explain why the liver can synthesize ketone bodies but cannot utilize it)

Q6) How can the blood gas analysis and urinary ketones be helpful in reaching the diagnosis? What is causing the elevation of plasma lactate in this patient; compare contrast with a patient with von Gierke disease?

Q7) List and explain the causes of the clinical hallmarks of von Gierke disease.

Q8) Why is the plasma free carnitine decreased while the acyl-conjugates (acyl-carnitine and acyl-glycine) are increased in the plasma and urine? (Discuss the source and function of carnitine; compare and contrast primary and secondary carnitine deficiencies)

Q9) Why is the clinical presentation in CPT-1 deficiency similar to MCAD but not similar to McArdle's disease?

Q10) Why are there increased dicarboxylic acids in the urine?

Q11) Why is there hyperammonemia? Why is hepatomegaly associated with elevated liver transaminases?

Q12) What therapeutic options are there for treating a patient with MCAD deficiency? Explain the rationale of therapy.

Additional Questions:

Q13) What is characteristic of β-oxidation in peroxisomes? Identify the biochemical defects in Refsum disease and Zellweger syndrome.

Q14) What is the metabolic fate of propionyl-CoA formed as a result of odd chain fatty acid oxidation in mitochondria? Indicate the vitamins required for its metabolism.

Q15) Identify the regulatory steps of fatty acid de-novo synthesis. How is it regulated?

Q16) Explain the central role of acetyl-CoA in metabolism. Indicate the subcellular location of each pathway.

Q17) Indicate sites of triacylglycerol (TAG) synthesis. What is the precursor of TAG synthesis in the intestine, in the liver and in the adipose tissue?

CASE 2 – Muscle Weakness

Modified from Higgins et al. Biochemistry

For the Medical Sciences

Objectives:

SOM. MK.I.BPM2.3.DM.2.BCHM.1385	Describe the regulation of blood glucose level in the fed state and fasting state
SOM. MK.I.BPM2.3.DM.2.BCHM.1386	Describe the glycogen synthesis and its regulation in the liver and muscle
SOM. MK.I.BPM2.3.DM.2.BCHM.1387	Describe glycogen degradation. Compare and contrast glycogen degradation in the muscle and the liver.
SOM. MK.I.BPM2.3.DM.2.BCHM.1388	Differentiate between the various glycogen storage disorders affecting the liver and muscle.
SOM. MK.I.BPM2.3.DM.2.BCHM.1389	Describe the role of hormones in the regulation of blood glucose level in the fed and fasting state.
SOM. MK.I.BPM2.3.DM.2.BCHM.1390	Describe the regulation of glycolysis in the liver and the muscle
SOM. MK.I.BPM2.3.DM.2.BCHM.1391	Review the carbohydrate metabolism lectures including concepts of signal transduction, enzyme regulation, glycolysis, gluconeogenesis, glycogen metabolism, glycogen storage disorders and regulation of blood glucose

Case:

In order to understand this case, students should be familiar with the material covered in the carbohydrate metabolism lectures: signal transduction, enzyme regulation, glycolysis, gluconeogenesis, glycogen metabolism, glycogen storage disorders and regulation of blood glucose.

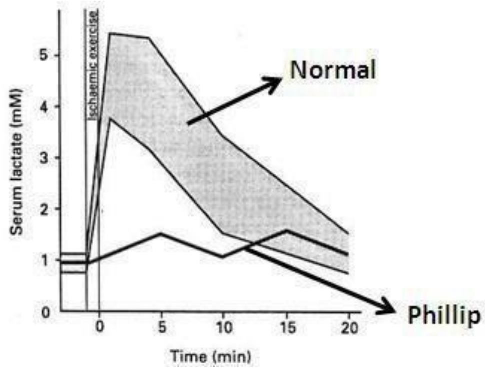
Introduction:

Skeletal muscle has a highly variable requirement for energy (in the form of ATP) depending on the demand placed on the muscle. Potentially, this energy can be derived from a variety of sources. However, during severe exercise when the energy demand is very high, blood is squeezed out of the muscle bed by contractile activity. The muscle is then temporarily starved of oxygen (becomes ischemic), making oxidative processes very difficult. In these circumstances, oxidation of fatty acids, normally the most efficient means of generating energy, cannot be relied upon to satisfy the large energy demand. Instead the muscle has to depend more heavily on anaerobic glycolysis. Even here the muscle faces problems since the diminished blood flow prevents glucose being imported to fuel glycolysis so the muscle's glycogen stores must be mobilized. In addition, the end product of anaerobic glycolysis, lactate, cannot be removed efficiently via the blood and has to accumulate within the muscle. As a consequence, humans cannot carry out severe ischemic exercise for more than a few minutes, as athletes will know only too well.

The Problem:

Phillip M., 16-year-old boy, sought medical help for progressive muscle weakness over a number of years. He experienced painful muscle cramps on severe exercise, but he could tolerate exercise normally. The effect of severe ischemic exercise on blood lactate is shown in figure 1.2. Severe exercise was followed by dramatically elevated serum of lactate dehydrogenase and creatine kinase-MM, which persisted for some hours after exercise had ceased. Phillip also reported that his exercise intolerance became worse after he had eaten a meal rich in starch.

Figure 1.2. Effect of ischaemic work of the forearm muscles on blood lactate. Data for the patient are shown (solid line) with results for normal subjects (shaded area).



Myoglobinuria was also present and there was evidence of mild hemolysis. Phillip's blood glucose level was normal and could be elevated by treatment with glucagon, it was decided to do a muscle biopsy. Table 1.1 shows the results of the muscle biopsy analysis.

Table 1.1. Analysis of metabolites in muscle biopsy samples.

	Tissue component			
Source of muscle	Glycogen (mg/g tissue)	Glucose 6-phosphate (μmol/g tissue)	Fructose 6-phosphate (μmol/g tissue)	Fructose 1,6-bisphosphate (μmol/g tissue)
Phillip	43.8	9.2	1.6	0.02
Normal volunteers (mean $\pm$ SD)	9.6 ± 1.8	0.5 ± 0.3	0.1 ± 0.05	0.61 ± 0.23

Q1) Why is lactate, rather than pyruvate, produced by normal muscle when it is working under ischemia?

Q2) What do the results of the ischemic exercise test suggest is wrong with the patient's muscles?

Q3) What is the normal effect of glucagon? What do the results of treating patient with glucagon indicate about his condition?

Q4) What can you conclude from the muscle biopsy analysis?

Q5) From your deductions so far, what do you think is the metabolic defect in this patient?

Q6) Why is the patient able to tolerate moderate, but not severe exercise?

Q7) What is the significance of his serum enzyme levels and the myoglobinuria?

Q8) Why is there an increased deposition of muscle glycogen (Table I.I)? Describe the normal glycogen structure.

Q9) What evidence may be present to support the suggestion of hemolysis? How does it arise in this patient?

Additional Questions:

Q10) In what ways is the structure of glycogen "abnormal" in some other glycogen storage disorders?

Q11) How is fructose 2,6 bisphosphate formed and what is its role in the liver? Describe 'feed-forward' and 'feedback' allosteric regulation of enzyme activity.

Q12) Describe regulation of enzyme activity by covalent modification, induction (activation) and repression (inhibition) with examples for each in carbohydrate metabolism.

Q13) Name another glycogen storage disorder affecting the muscle. Indicate the enzyme deficiency in the disorder. Why is hypoglycemia not a feature of this disorder?

Q14) Recapitulate the reactions of glycogenolysis and glycogenesis. Explain how these pathways are reciprocally regulated by glucagon, epinephrine and insulin. Identify the role of cAMP in glycogen metabolism

Q15) Name other enzyme deficiency disorders that can result in hemolysis. Identify the cause of hemolysis in each of these disorders.